Statement of Purpose for Ph.D. in Computer Science

I am interested in pursuing a Ph.D. in applied cryptography and systems security. I want to build practical systems that use cryptography for precise security while preserving privacy. In particular, I am excited to discover what stronger privacy we can achieve when we deprioritize backward compatibility. Security needs and available tools develop fast, and systems that commit to current tools risk growing obsolete. I want to instead tackle long-term security problems with clean-slate solutions that can efficiently scale and evolve for changing system needs. In my previous research, I developed a sparse polynomial commitment scheme from lattices, faster fully homomorphic encryption in deep circuits, and a privacy-preserving authentication logging protocol. I used special-purpose cryptographic primitives to achieve high-performance, which gives me confidence that I can solve real-world problems with a clean-slate approach.

Research in Cryptography. In 2024, I worked with Professor Dan Boneh to build the first sparse polynomial commitment scheme (PCS) from lattices. PCS are a primitive that allows one to commit to a polynomial and later prove its evaluations. A 'sparse' PCS enables efficient commitments to large structures, that would otherwise be too expensive, by leveraging the sparsity of data. We adapted an existing field-based sparse PCS called Spark [7] to work on rings and achieved optimal prover costs that depend only on the sparsity of the committed polynomial. We standardized the scheme to be able to use any (non-sparse) PCS for lattices in the commit phase while maintaining optimal prover costs. The main challenge was achieving soundness (meaning the PCS verifier rejects an invalid evaluation proof with high probability) after transitioning Spark to rings, because there exist complicating factors like zero divisors in rings that are not accounted for in prior field-based constructions. I proposed to use sum-checks over rings to achieve soundness, by sampling randomness from domains that avoid zero divisors. Additionally, I proved the soundness of a core technique in our PCS that checks whether two multisets contain the same ring elements—a result previously only known for field elements. To optimize prover costs, I adapted Spark's time-optimal memory-checking algorithm to rings, including a new enumeration of ring elements over a boolean hypercube. We will be submitting our paper (available on my website) to cryptography conferences soon.

I am most interested in research with a strong real-world motivation and sought more applied cryptography projects. Fully homomorphic encryption (FHE) [4] is a privacy-enhancing tool that allows computing on encrypted data without revealing the underlying plaintext. Many FHE applications work with convolutional neural networks (CNNs) but they treat CNNs designed for unencrypted data as black boxes, simply feeding them FHE-encrypted data. This naive approach raises two problems: (1) CNNs require many multiplications, which, while trivial in standard settings, become a bottleneck in homomorphic settings, and (2) repeated computations can overflow plaintexts, hurting data integrity. In 2023, advised by Professor John Mitchell, I addressed problem (1) by developing 'Swap', a method that transforms any traditional CNN to achieve faster encrypted inference when working on FHE-encrypted data. Swap performs less ciphertext multiplications by averaging data before applying activation functions, while maintaining accuracy by selecting where to apply the technique. I applied Swap to the Lenet-5 neural network in the FHE setting for a 27% speedup with 90% accuracy. I addressed problem (2) by developing the first plaintext overflow detection protocol in fixed-point FHE. I showed its applicability to the Cheon-Kim-Kim-Song [2] and Brakerski-Gentry-Vaikuntanathan [1] schemes and proved its indistinguishability under chosen plaintext attack. I solo-published and presented this work [5] at the 14th ACM Conference on Data and Application Security and Privacy. However, now, rather than optimizing existing tools, I wanted to tackle more challenging projects that introduce entirely new security capabilities.

Research in Systems. Authentication logging is a way to detect credential compromise

by recording a user's complete authentication history. However, many existing single sign-on schemes like "Sign in with Google" achieve universal logging by compromising user privacy and giving a third party access to a user's authentication history across all her accounts. The current state-of-the-art solution for privacy-preserving authentication logging is larch [3], but it has two limitations: (1) it requires a trusted party to run the log server at enrollment and audit times for correct service, and (2) it loses user privacy when the log server colludes with a relying party (RP). Observing that these limitations partly arise from larch's goal to achieve backward compatibility, I proposed a clean-slate solution. Since February, I have worked with Professor David Mazières on the system's top-level design, where users' encrypted authentication records are posted on a public blockchain with a valid proof, and an RP authorizes log-in after checking that the correct content has been logged. Collaborating with Professor Emma Dauterman who authored larch, I devised cryptographic primitives that ensure even an attacker that controls both the log server and RP cannot link a user across accounts. I eradicated the need for a trusted log server at audit time by leveraging proofs of inclusion in a public blockchain. I contributed to minimizing load on involved parties by replacing two heavyweight zkSNARKs in our initial design (with an 8-second latency) with Sigma protocols and blind signatures. My Rust implementation of the final protocol is expected to complete authentication in 1 second on four CPU cores, which is comparable to regular log-in processes with no secure logging.

Outside of cryptographic systems, I saw how solutions that are independent of today's software and hardware can achieve better scalability, when I developed such a Transport Layer Security (TLS) client identification tool with Professor Zakir Durumeric in 2023. Client identification infers a client's identity by observing its network behavior and is essential for investigating anomalous clients. However, I showed that three state-of-the-art tools can collectively identify only 6% of clients in captured traffic, due to their reliance on hardcoded databases that grow outdated as the network landscape changes. Instead, I built and evaluated Clid [6]: a self-sustaining TLS client identification tool that uses unsupervised learning and optimization to identify domain names that are most indicative of the clients' identity, among their previous connections. Clid maps at least 60% of clients in all test TLS sets to informative domain names.

My clean-slate approach has taught me how to build systems that operate 'the right way', rather than conforming to arbitrary constraints. Now, I want to discover how to maintain the benefits of this approach in systems that compose cleanly with existing applications. Therefore, Professor Emma Dauterman's work on usable cryptographic systems appeals to me, especially in private query or user data protection where high-performance is asked. One direction I want to explore is empowering honest users to achieve security without assuming non-colluding parties or special hardware. For instance, our authentication protocol tolerates some log server misbehavior at authentication time, as users can prove the misbehavior later. Extending this, I want to discover if self-policing users can help build systems that execute correctly even with malicious centralized parties. Another approach I am interested in is advancing computation on encrypted data through such means as encrypted graphic processing units, extending my FHE project. I aspire to work with Professor Dan Boneh to build *precise* cryptographic tools that minimize system burden—for instance, a new proof system that can evaluate arbitrary numbers of batch proofs in split-trust settings, where multiple parties need to exchange proofs, but each only has selective knowledge. A challenge in my authentication logging work was building a central service without making it a single point of vulnerability. For instance, after one account belonging to a user is compromised, preventing the user from losing access to the logging service she uses for *all* her accounts is non-trivial. With Professor David Mazières, I want to explore using distributed systems to compartmentalize risks of account compromise, while maintaining a centralized service for good usability.

References

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